

Lectures in Mathematical Modeling

Supplementary 8: Probability Essentials for Modeling and Simulation

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Learning goals

By the end of this two-hour supplementary lesson, students should be able to:

- describe a random experiment using a **sample space**, **event**, and probability model;
- compute simple probabilities using counting, complements, and inclusion–exclusion;
- distinguish **conditional probability**, **independence**, and disjointness;
- use **Bayes’ theorem** to update probabilities from evidence;
- model uncertainty using **random variables**, PMF/PDF, CDF, expectation, and variance;
- recognise when binomial, Poisson, exponential, normal, or Markov-chain models are appropriate;
- explain how the Law of Large Numbers and Central Limit Theorem justify Monte Carlo simulation.

1 Why probability belongs in mathematical modeling

A deterministic model gives one output when the input is fixed. Real systems often require a different language. Customers do not arrive exactly every five minutes; a sensor reading contains noise; a production line occasionally creates defective products; weather changes from day to day. A modeling report should therefore be able to say not only “what will happen”, but also “how likely”, “with what variability”, and “how confident are we?”

Launch activity: deterministic or stochastic?

Classify each situation as mainly deterministic, stochastic, or mixed.

1. A robot travels 120 m at a constant speed of 1.5 m/s.
2. Customers enter a canteen during lunch time.
3. A component has a 2% chance of failing during one month.
4. A delivery route is planned, but travel time depends on weather and crowding.

A strong modeling answer usually separates the predictable structure from the random variation.

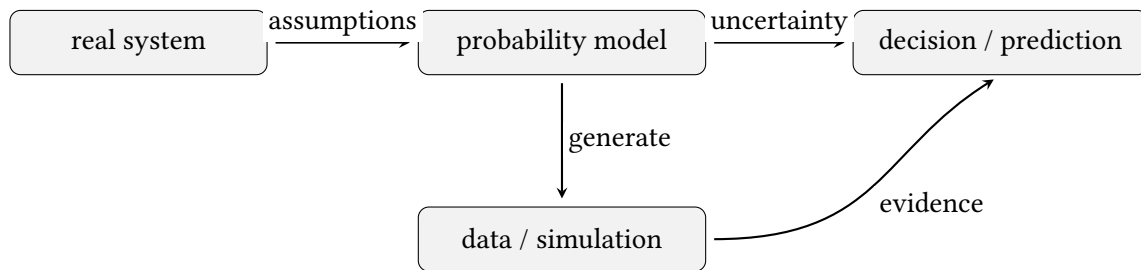


Figure 1: Probability enters modeling when assumptions must describe uncertainty, variability, and evidence rather than a single fixed outcome.

2 Sample spaces, events, and counting

Sample space and event

A **random experiment** is a procedure whose outcome is uncertain but whose possible outcomes can be described. The **sample space** Ω is the set of all possible outcomes. An **event** is a subset $A \subseteq \Omega$.

Worked example 1: two dice

For rolling two fair dice,

$$\Omega = \{(i, j) : 1 \leq i, j \leq 6\}, \quad |\Omega| = 36.$$

The event “sum is 7” is

$$\{(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)\},$$

so $\mathbb{P}(\text{sum} = 7) = 6/36 = 1/6$.

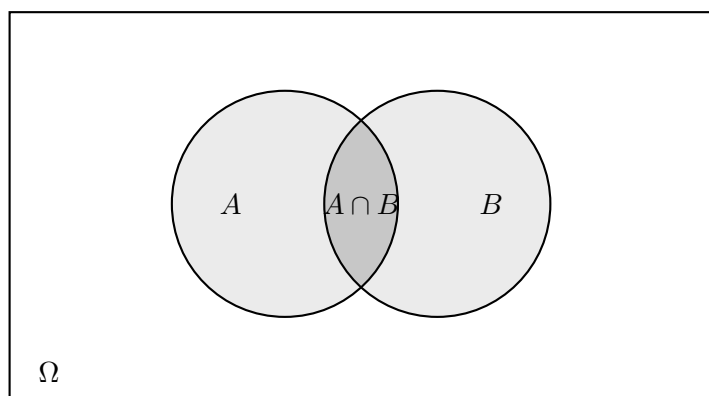
Probability rules

For events A and B ,

$$\mathbb{P}(A^c) = 1 - \mathbb{P}(A), \quad \mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B).$$

If all outcomes are equally likely and Ω is finite, then

$$\mathbb{P}(A) = \frac{|A|}{|\Omega|}.$$



$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$$

Figure 2: Inclusion–exclusion subtracts the overlap once because it is counted in both A and B .

Worked example 2: birthday problem

Assume birthdays are uniformly distributed over 365 days. For $n = 23$ people, compute the probability that at least two people share a birthday.

It is easier to use the complement. The probability all birthdays are distinct is

$$\frac{365 \cdot 364 \cdot 363 \cdots (365 - 22)}{365^{23}} \approx 0.4927.$$

Therefore

$$\mathbb{P}(\text{at least one shared birthday}) \approx 1 - 0.4927 = 0.5073.$$

Only 23 people are enough for a probability slightly above one half.

Checkpoint: Checkpoint

A bag has 5 red balls and 3 blue balls. Two balls are drawn without replacement. Is “first ball red” independent of “second ball red”? Explain using conditional probability.

3 Conditional probability, independence, and Bayes

Conditional probability

For events A and B with $\mathbb{P}(B) > 0$,

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

This means: restrict the sample space to B , then renormalise.

Independence

Events A and B are **independent** if

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B),$$

or equivalently $\mathbb{P}(A \mid B) = \mathbb{P}(A)$ when $\mathbb{P}(B) > 0$. Independence is not the same as disjointness. If two positive-probability events are disjoint, knowing one happened makes the other impossible, so they are dependent.

Total probability and Bayes’ theorem

If $B_1, \dots, B_n$ partition Ω , then

$$\mathbb{P}(A) = \sum_{i=1}^n \mathbb{P}(A \mid B_i)\mathbb{P}(B_i).$$

Bayes’ theorem updates the probability of a cause after observing evidence:

$$\mathbb{P}(B_k \mid A) = \frac{\mathbb{P}(A \mid B_k)\mathbb{P}(B_k)}{\sum_{i=1}^n \mathbb{P}(A \mid B_i)\mathbb{P}(B_i)}.$$

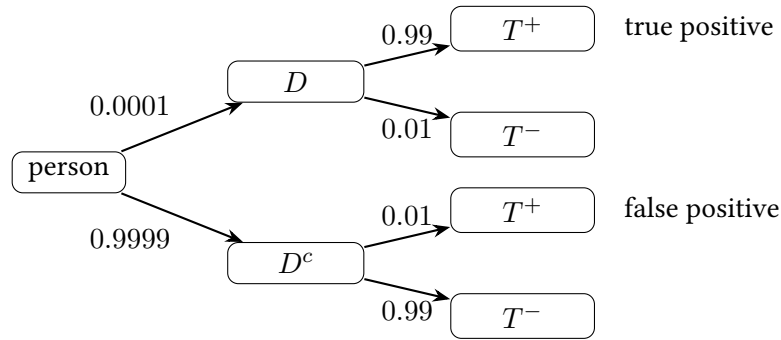


Figure 3: A probability tree separates prior probability from test reliability. The false-positive branch can dominate when the base rate is very small.

Worked example 3: medical testing and base rate

A disease affects 1 in 10,000 people. A test has 99% sensitivity and 99% specificity. A randomly selected person tests positive. What is the probability they have the disease?

Let D be “has disease” and T^+ be “positive test”. Then

$$\mathbb{P}(D) = 10^{-4}, \quad \mathbb{P}(T^+ | D) = 0.99, \quad \mathbb{P}(T^+ | D^c) = 0.01.$$

Therefore

$$\mathbb{P}(D | T^+) = \frac{0.99(10^{-4})}{0.99(10^{-4}) + 0.01(0.9999)} \approx 0.0098.$$

Even after a positive test, the probability is under 1%. This is not because the test is useless; it is because the disease is rare.

Model criticism: Model criticism: Bayes in reports

When using Bayes’ theorem, a report should state the prior probabilities, evidence reliability, and what changes after observing evidence. Many wrong conclusions come from ignoring the base rate.

4 Random variables, distributions, expectation, and variance

Random variable

A **random variable** is a function $X : \Omega \rightarrow \mathbb{R}$ that assigns a numerical value to each outcome. It is **discrete** if its possible values are finite or countable, and **continuous** if probabilities are described by intervals and integrals.

PMF, PDF, and CDF

For a discrete random variable, the **probability mass function** is

$$p_X(x) = \mathbb{P}(X = x), \quad \sum_x p_X(x) = 1.$$

For a continuous random variable, the **probability density function** satisfies

$$\mathbb{P}(a \leq X \leq b) = \int_a^b f_X(x) dx.$$

For any random variable, the **cumulative distribution function** is

$$F_X(x) = \mathbb{P}(X \leq x).$$

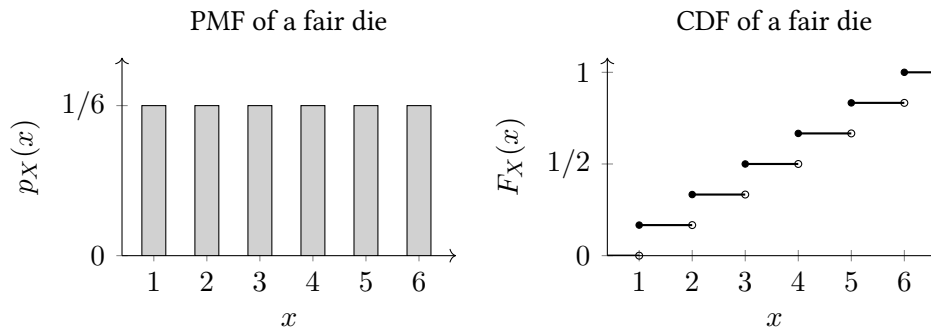


Figure 4: A discrete distribution has mass at points. Its CDF increases in jumps.

Expectation and variance

The **expected value** is the long-run average:

$$\mathbb{E}[X] = \sum_x x p_X(x) \quad (\text{discrete}), \quad \mathbb{E}[X] = \int_{-\infty}^{\infty} x f_X(x) dx \quad (\text{continuous}).$$

The **variance** measures spread:

$$\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2.$$

The standard deviation is $\text{SD}(X) = \sqrt{\text{Var}(X)}$.

Worked example 4: die roll mean and variance

Let X be the result of a fair six-sided die. Then

$$\mathbb{E}[X] = \frac{1 + 2 + 3 + 4 + 5 + 6}{6} = 3.5,$$

$$\mathbb{E}[X^2] = \frac{1 + 4 + 9 + 16 + 25 + 36}{6} = \frac{91}{6},$$

so

$$\text{Var}(X) = \frac{91}{6} - (3.5)^2 = \frac{35}{12} \approx 2.917.$$

The standard deviation is about 1.708.

Worked example 5: expected profit

An insurer sells a one-year policy for a premium of \$1,500. The claim is \$200,000 if death occurs, and the probability of death in the year is $p = 0.01$. The insurer's expected profit is

$$0.99(1500) + 0.01(1500 - 200000) = 1485 - 1985 = -500.$$

At this premium the policy loses \$500 in expectation. This connects directly to decision theory and risk modeling.

5 Named distributions for modeling

When to use which distribution

Distribution	Modeling question	Key quantities
Bernoulli(p)	one success/failure trial	mean p , variance $p(1 - p)$
Binomial(n, p)	number of successes in n independent trials	mean np , variance $np(1 - p)$
Poisson(λ)	number of random arrivals in a fixed interval	mean λ , variance λ
Exponential(λ)	waiting time between Poisson arrivals	mean $1/\lambda$, memoryless
Normal(μ, σ^2)	measurement error, averages, natural variability	mean μ , variance σ^2

Worked example 6: binomial quality control

A factory has defect rate $p = 0.02$. In a batch of $n = 100$ items, let X be the number of defective items. Then $X \sim \text{Bin}(100, 0.02)$ and

$$\mathbb{E}[X] = 100(0.02) = 2.$$

The probability of at most 2 defects is

$$\mathbb{P}(X \leq 2) = \sum_{k=0}^2 \binom{100}{k} (0.02)^k (0.98)^{100-k} \approx 0.677.$$

Worked example 7: Poisson arrivals and exponential waiting

Customers arrive at a service window at average rate 12 per hour. If arrivals are modeled as Poisson, then the number of arrivals in one hour is $X \sim \text{Pois}(12)$:

$$\mathbb{P}(X = 10) = e^{-12} \frac{12^{10}}{10!} \approx 0.105.$$

The waiting time until the next customer is $T \sim \text{Exp}(12)$ hours, so

$$\mathbb{E}[T] = \frac{1}{12} \text{ hour} = 5 \text{ minutes}.$$

This is the probability foundation of queue simulation.

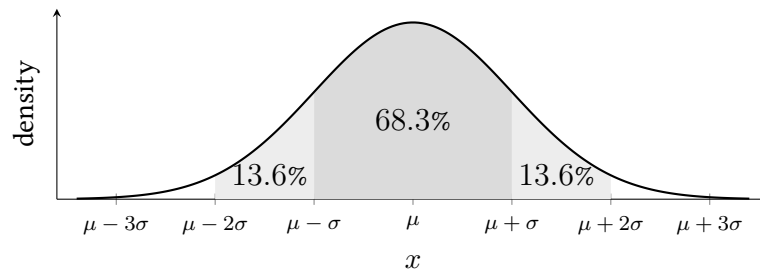


Figure 5: The normal distribution is used for measurement error, natural variation, and sampling distributions. About 68.3% of values lie within one standard deviation of the mean.

Checkpoint: Checkpoint

Which distribution would you use for each case?

1. Number of students arriving in the next 10 minutes.
2. Whether a randomly selected component is defective.
3. Waiting time until the next customer.
4. Measurement error of repeated sensor readings.

6 Monte Carlo, LLN, and CLT

The link between probability and simulation is simple: if we cannot compute a quantity exactly, we can sometimes estimate it by repeated random trials. The theory tells us why the estimate improves.

Law of Large Numbers

Let $X_1, X_2, \dots$ be independent and identically distributed with mean μ . The sample mean

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$$

converges to μ in probability. In modeling language: repeated simulation averages approach the expected value.

Central Limit Theorem

If $X_1, \dots, X_n$ are i.i.d. with mean μ and variance σ^2 , then for large n ,

$$\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \approx Z \sim \mathcal{N}(0, 1).$$

Therefore an approximate 95% confidence interval for μ is

$$\bar{X}_n \pm 1.96 \frac{s}{\sqrt{n}},$$

where s is the sample standard deviation.

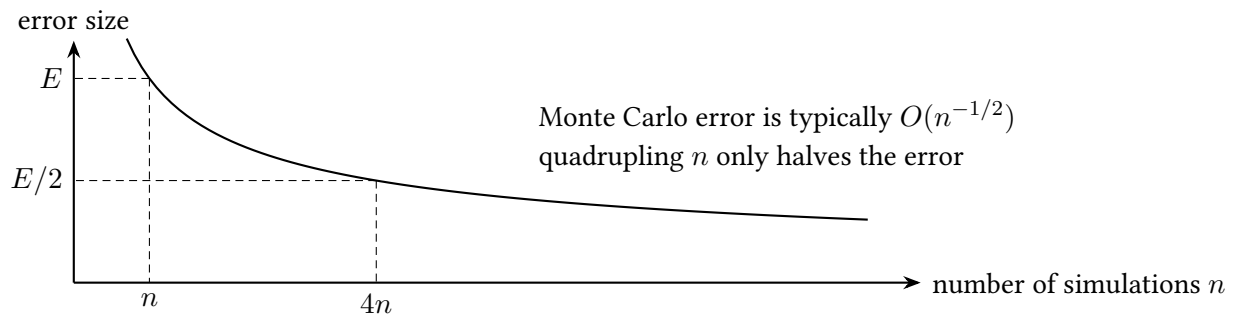


Figure 6: The $1/\sqrt{n}$ convergence rate explains why Monte Carlo is flexible but often slow.

Worked example 8: sample size for Monte Carlo

Suppose a Monte Carlo estimator is an average of Bernoulli indicators, so its standard deviation is at most $1/2$. How large should n be for a normal-approximation 95% confidence interval half-width at most 0.005?

$$1.96 \frac{1/2}{\sqrt{n}} \leq 0.005 \implies \sqrt{n} \geq 196 \implies n \geq 38416.$$

Thus the normal-approximation design requires at least 38416 samples under this worst-case variance bound. Finite-sample coverage should be checked separately when the estimated probability is near 0 or 1.

Model criticism: Reporting simulation uncertainty

A competition report should not only give a simulated estimate. It should also report the number of replications, the standard error, a confidence interval, and whether the conclusion changes under reasonable parameter perturbations.

7 A first Markov-chain model

Discrete-time Markov chain

A sequence $X_0, X_1, X_2, \dots$ on a finite state space is a **Markov chain** if the next state depends on the present state but not on the full past:

$$\mathbb{P}(X_{n+1} = j \mid X_n = i, X_{n-1}, \dots, X_0) = \mathbb{P}(X_{n+1} = j \mid X_n = i) = P_{ij}.$$

The matrix $P = (P_{ij})$ is the **transition matrix**; each row sums to 1.

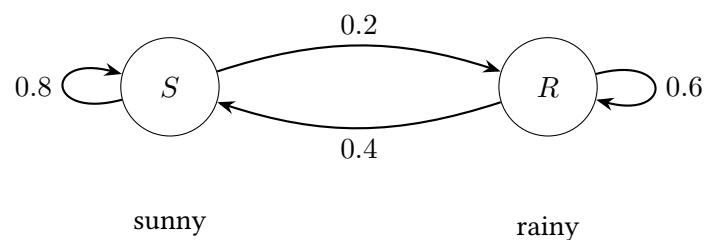


Figure 7: A two-state Markov chain for weather. The probabilities leaving each state add to 1.

Worked example 9: two-day weather prediction

States are S = sunny and R = rainy. Suppose

$$P = \begin{pmatrix} 0.8 & 0.2 \\ 0.4 & 0.6 \end{pmatrix},$$

where rows are today's weather and columns are tomorrow's weather. If today is sunny, the distribution after two days is

$$(1, 0)P^2 = (1, 0) \begin{pmatrix} 0.72 & 0.28 \\ 0.56 & 0.44 \end{pmatrix} = (0.72, 0.28).$$

Thus the probability of rain two days from now is 0.28.

Worked example 10: stationary distribution

A stationary distribution $\pi = (\pi_S, \pi_R)$ satisfies $\pi P = \pi$ and $\pi_S + \pi_R = 1$. From the weather matrix,

$$\pi_S = 0.8\pi_S + 0.4\pi_R \implies 0.2\pi_S = 0.4\pi_R \implies \pi_S = 2\pi_R.$$

Together with $\pi_S + \pi_R = 1$, we get

$$\pi = (2/3, 1/3).$$

In the long run, about two-thirds of days are sunny.

8 Differentiated modeling studio

Studio task: canteen queue uncertainty

A canteen receives students at an average rate of 18 per 10 minutes. During peak time, arrivals are modeled as Poisson.

1. Compute the probability of exactly 20 arrivals in the next 10 minutes.
2. Compute or approximate the probability of more than 24 arrivals.
3. If one server can handle 20 students per 10 minutes, what risk does this create?
4. Write one paragraph explaining whether the Poisson assumption is reasonable.

Competition-style reporting paragraph

We model arrivals in each 10-minute period as $X \sim \text{Pois}(18)$, assuming independent arrivals at an approximately constant rate during the short peak interval. This gives $\mathbb{E}[X] = 18$ and $\text{Var}(X) = 18$, so the typical fluctuation is about $\sqrt{18} \approx 4.24$ students. Since the server capacity is 20 students per 10 minutes, the system may become overloaded in a non-negligible fraction of intervals. We therefore recommend evaluating service capacity using not only the mean arrival rate, but also the upper tail probability $\mathbb{P}(X > 20)$ and sensitivity to the estimated rate.

Model criticism: Model criticism

Probability models are assumptions, not facts. A Poisson arrival model assumes independent arrivals and a roughly constant rate. If students arrive in groups after a bell rings, the independence

assumption is weak. If the rate changes sharply during lunch, the constant-rate assumption is weak. Validation should compare both the mean and the variability of observed arrivals against the model.

9 Core practice and extension problems

Core practice

1. Two fair dice are rolled. Find $\mathbb{P}(\text{sum} = 8)$ and $\mathbb{P}(\text{sum} \geq 10)$.
2. A bag contains 5 red and 3 blue balls. Two are drawn without replacement. Find the probability both are red.
3. A factory has two machines. Machine A produces 70% of items with defect rate 1%; machine B produces 30% with defect rate 4%. A randomly chosen item is defective. Find the probability it came from B.
4. Let X be the number of heads in 4 tosses of a fair coin. Write its distribution and compute $\mathbb{E}[X]$.
5. If $X \sim \text{Pois}(5)$, compute $\mathbb{P}(X = 3)$ and $\mathbb{P}(X = 0)$.
6. If $T \sim \text{Exp}(6)$ hours, find $\mathbb{E}[T]$ and $\mathbb{P}(T > 0.5)$.

Practice check and hints

1. The probabilities are $5/36$ and $1/6$.
2. The probability is $\binom{5}{2}/\binom{8}{2} = 5/14$.
3. Bayes' rule gives $0.30(0.04)/[0.70(0.01) + 0.30(0.04)] = 12/19$.
4. $X \sim \text{Bin}(4, 1/2)$, so $\mathbb{P}(X = k) = \binom{4}{k}/16$ and $\mathbb{E}[X] = 2$.
5. The probabilities are $e^{-5}5^3/3! \approx 0.14037$ and $e^{-5} \approx 0.006738$.
6. $\mathbb{E}[T] = 1/6$ hour = 10 minutes, and $\mathbb{P}(T > 0.5) = e^{-3} \approx 0.0498$.

Modeling challenge

A small help desk receives requests according to a Poisson process with rate $\lambda = 10$ per hour. Service time is approximately exponential with mean 5 minutes.

1. Convert the mean service time into a service rate μ per hour.
2. Compute the utilisation ratio $\rho = \lambda/\mu$.
3. Explain what it means if ρ is close to 1.
4. Suggest one data source that could validate the arrival-rate assumption and one source that could validate the service-time assumption.

Modeling checks

The service rate is $\mu = 12$ per hour and $\rho = 10/12 = 5/6 \approx 0.833$. Interpret this as a fitted load indicator, not proof that Poisson arrivals or exponential service times are valid. A complete validation plan specifies timestamped arrival data, observed service durations, diagnostics for time variation and dependence, and a held-out comparison.

Extension practice

1. Prove that if A and B are independent, then A and B^c are independent.
2. Let $X \sim U(0, 1)$ and $Y = -\ln X$. Derive the CDF of Y and identify its distribution.
3. Let $X_1, \dots, X_n$ be independent Bernoulli(p). Use linearity of expectation to show $\mathbb{E}[\sum X_i] = np$.
4. In the weather Markov chain, if today is rainy, compute the probability of a sunny day three days from now.
5. Give an example where $\text{Cov}(X, Y) = 0$ but X and Y are not independent.

Extension hints

Use $\mathbb{P}(A \cap B^c) = \mathbb{P}(A) - \mathbb{P}(A \cap B)$ in Question 1. In Question 2, $F_Y(y) = 1 - e^{-y}$ for $y \geq 0$, so $Y \sim \text{Exp}(1)$. Question 3 follows by summing expectations. The three-day sunny probability in Question 4 is 0.624. For Question 5, one example is $X \sim U(-1, 1)$ with $Y = X^2$: the covariance is 0, but the variables are dependent.

Model audit card

Audit before trusting or communicating a model			
Purpose	What question or decision does the model support?	Data	Which inputs are observed, derived, or simulated?
Assumptions	Which simplifications may change the answer?	Units	Are dimensions and scales consistent?
Verification	Were equations, code, constraints, and limiting cases checked?	Validation	Is performance tested against independent evidence?
Uncertainty	How do measurement, parameter, solver, and structural errors affect conclusions?	Ethics	Who benefits, who bears risk, and what fairness criterion matters?
Reproducibility	Are data provenance, versions, seeds, and transformations recorded?	Communication	Are claims conditional, traceable, and decision-relevant?

Further reading

- Introductory Statistics 2e (OpenStax)

Sources, provenance, and licence

These notes follow the iterative modeling and assessment perspective in the SIAM/COMAP GAIMME report. Competition-specific remarks are a snapshot checked on 20 August 2026 against the official COMAP HiMCM instructions; readers should verify current rules before submission. Unless a source is explicitly identified, classroom datasets and numerical scenarios are synthetic or simulated teaching examples and are not empirical claims about a real institution. Third-party references retain their own terms. Original teaching material by Kenneth Sok Kin Cheng is released under CC BY-NC-SA 4.0.

Student response guide and checking criteria
For closed calculations, show the definition, substitution, units, and at least one independent check; the worked examples in this unit are method hints, but you must complete the arithmetic yourself. Open modeling responses are checked on five dimensions: a clear purpose and output; stated data and provenance; defensible assumptions and scope; traceable computational verification and model validation; and alignment among uncertainty, limitations, and the recommendation. Support each dimension with concrete evidence rather than writing only “completed.”